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SOIL PHYSICS

Preferential Flow in Soils: Review of Role in Soil Carbon Dynamics, Assessment of Characteristics, and Performance in Ecosystems

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Abstract—Rapid and unstable preferential flow has a significant impact on soil carbon cycle. This review aims to explore the effects of preferential flow on the soil carbon cycle and indicate its characteristics and ecological responses in different ecosystems. This study concluded that preferential flow influences soil carbon cycle through various mechanisms, such as facilitating rapid transport of dissolved organic matter, shaping the distribution and aggregation patterns of soil organic carbon, and enhancing soil microbial activity and organic matter decomposition. The characteristics of preferential flow include surrounding characteristics, rapid non-equilibrium infiltration characteristics, fluctuating characteristics, universal characteristics, lateral infiltration characteristics. Those characteristics could also affect the spatial distribution of soil organic carbon. In addition, this review examines the phenomenon of preferential flow in farmland, forest, wetland, desert, and permafrost ecosystems. Finally, we provide insightful perspectives on future research directions, emphasizing the importance of advancing our understanding of preferential flow mechanisms. It also serves as a valuable resource for future research aimed at unraveling the underlying mechanisms of preferential flow and developing effective soil carbon management strategies.

Keywords: soil matrix, solute transport, ecosystems effects, root channels, soil carbon cycle

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INTRODUCTION

Population growth and extreme weather events have exacerbated global water scarcity, intensifying the conflict between water supply and human demand. In recent years, frequent extreme rainfall events and changes in rainfall intensity have disrupted surface runoff distribution and groundwater recharge, thereby altering vegetation spatial patterns and exacerbating soil aridity [65]. Soil water flow serves as a crucial link between surface runoff and groundwater [55]. The rapid water infiltration through preferential flow paths, such as soil macropores, to reach the groundwater layer has significant implications for regional and global water cycles [27]. Preferential flow, as a common phenomenon in pedological perspectives, has been paid more attentions in soil science and environmental science. While different research areas offer distinct definitions of preferential flow, it is generally agreed to be the phenomenon of rapid unsteady flow of water and solutes along specific paths in heterogeneous soils [2]. The occurrence and effects of preferential flow are widespread, particularly in different ecosystems. For example, as an important part of terrestrial ecosystems, forest catchments hold most of the water resources available for human use [37], and forest ecosystems have important ecological services such as soil and water conservation. Soil moisture movement can regulate the flow state and interaction between groundwater and surface water [55], which plays an important role in the ecohydrological cycle. Therefore, the study of preferential flow has been the focus of recent years.

Preferential flow has both positive and negative impacts in different ecosystems. Its positive impacts primarily include: (1) reducing surface runoff generation through preferential flow paths [68], particularly via dead root channels resulting from root decay, which facilitate water infiltration [1, 30]; (2) accelerating solute transport, which enhances soil nutrient cycling and promotes the retention of soluble organic carbon and root secretions through interactions with soil particles surrounding the preferential flow paths [21]. The nutrient cycling in the root zones also benefits from the constraints imposed by preferential flow intensity [9]; (3) the response of preferential flow to precipitation affects deep groundwater replenishment in the soil and influences the quantity and quality of regional water

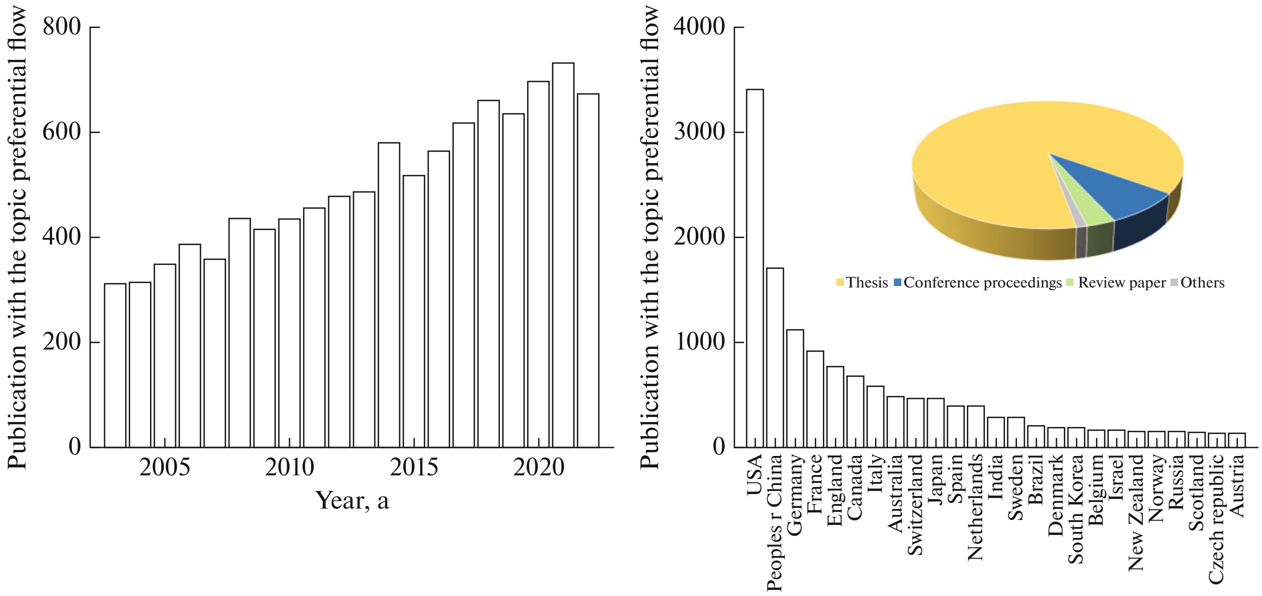

Fig. 1. The development of preferential flow research in the last 20 years from Web of Science.

resources [32]. On the other hand, the negative impacts of preferential flow mainly encompass: (1) the rapid transport of pesticides and fertilizers from agricultural ecosystems through preferential flow, reducing direct interaction with the soil matrix [64] and causing groundwater pollution; (2) the leaching of nutrients from the inner wall of preferential flow paths, leading to nutrient losses up to 2.5 times higher than under conditions without preferential flow paths [77], resulting in soil fertility decline and affecting vegetation [41]; (3) preferential flow can alter soil pore water pressure, reduce soil slope stability, induce slope instability, and increase the likelihood of geological disasters [12]. Preferential flow influences material circulation, information transfer, energy flow, and solute transport. The preferential flow paths intricately link soil water, surface water, and groundwater, consequently influencing ecohydrological processes at various spatial scales.

The review summarizes and analyzes the quantity, types, and countries of publication for preferential flow research papers in the past 20 years, as shown in Fig. 1. The analysis reveals a promising increase in the research dedicated to understanding preferential flow in recent years. Moreover, the research highlights the extensive distribution of studies on this subject, with a substantial body of research originating from the United States (31.41%), China (15.72%), and Germany (10.31%). Notably, a significant proportion of the scholarly output pertaining to preferential flow in the past two decades comprises thesis contributions (87.78%). These findings indicate a growing interest in the preferential flow. Meanwhile, more studies have found that preferential flow has an important role in soil carbon dynamics. This review explored the role of preferential flow on soil carbon sequestration and systematically concluded the preferential flow characteristics and their performance in different ecosystems. These preferential flow characteristics indirectly reveal its intensity and effectively demonstrate its ability to recharge groundwater and maintain above-ground vegetation and influence the soil carbon cycle, thus enhancing the comprehensive understanding of the effects of preferential flow on soil carbon dynamic.

Effects of Preferential Flow on Soil Carbon Dynamics

The macropores within the soils affect the preferential transport of soil water and solute. In general, soil pore networks can be classified into three categories based on different scales, such as the main channels larger than 300 μm, the second channels being 30–300 μm, and the third channels smaller than 30 μm (Fig. 2) [8, 31]. With respect to pores larger than 300 μm as the main channels for preferential water transport at the pore scale, they are able to transport more water and dissolved organic carbon through the channels [87]. For pores being 30–300 μm, they have the greatest impact on organic matter decomposition and soil respiration, and also have an important role in microbial carbon handling [3]. The pores smaller than 30 μm are more suitable for storing organic matter.

The relationship between preferential flow and the soil carbon cycle constitutes a significant subject in soil science. Within the soils, heterogeneous soil macropore networks as preferential flow paths facilitate water flow and solute transport, thereby influencing soil carbon cycle. Firstly, preferential flow exerts a substantial impact on the transport and distribution of soil organic carbon. The persistence and reactivation

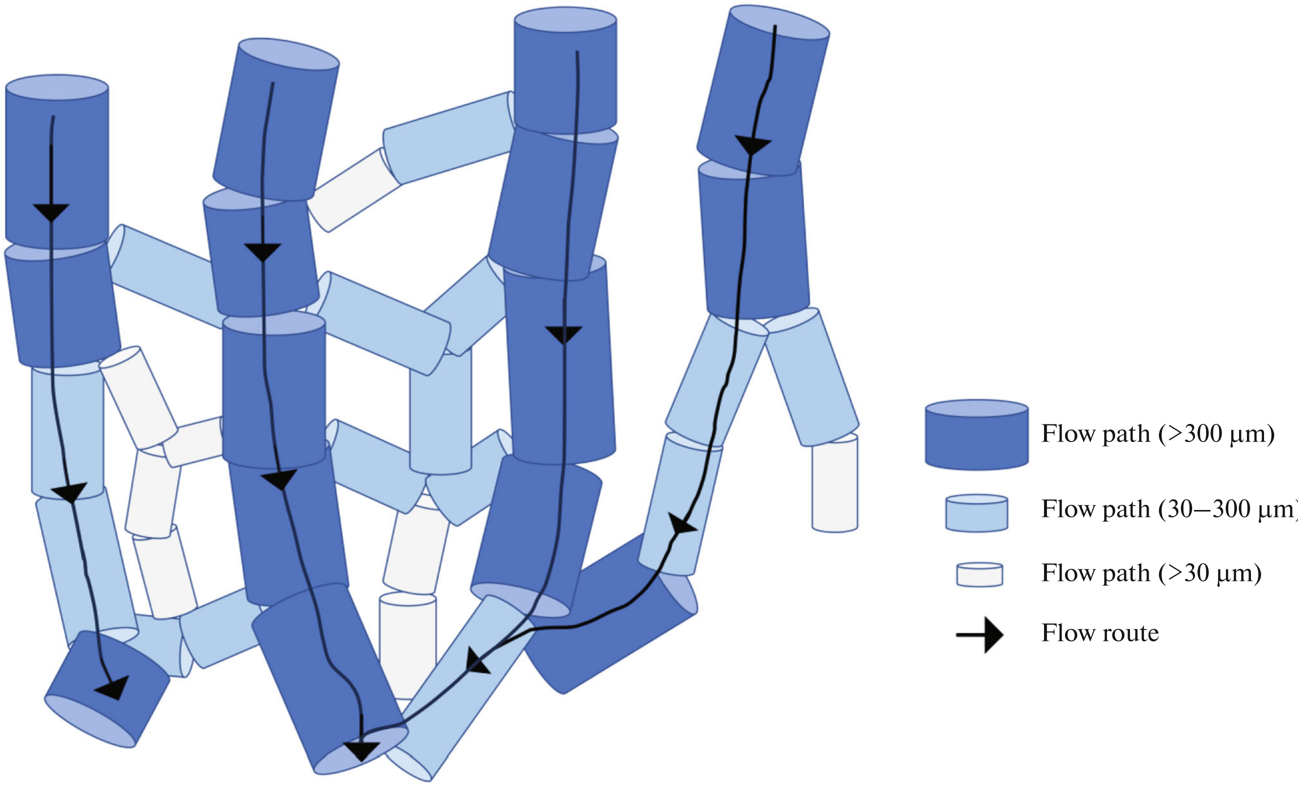

Fig. 2. The conceptual model shows the relationship between pore size, water flow in the preferential flow path. Adapted from Franklin et al. [19].

of stable preferential flow paths contribute to the localized accumulation of soil matrix nutrients within these paths, ultimately influencing soil biogeochemistry and carbon cycling [17]. Preferential flow channels are typically formed by larger pore spaces and more permeable soil structures. And the flow rates of preferential water are higher. As a result, preferential flow paths contribute to the rapid transport of dissolved organic matter to relatively deeper depths in the soil. Consequently, soil carbon can be transported to deeper soil layers or even groundwater, leading to spatially uneven aggregation and distribution of soil organic carbon [6]. Secondly, preferential flow potentially enhances the activity of soil microorganisms and enzymes [18]. Compared to soil matrix, preferential flow paths have higher content of soil organic matter and higher turnover of N, which may be more readily available by microorganisms, increasing microbial uptake of nutrients [20, 24]. Microorganisms may release carbon dioxide through the degradation of organic matter and subsequent respiration [33], while enzymes participate in the degradation and transformation processes of organic matter. Plant root channels and specific macropores formed by soil animals are notable contributors to preferential flow paths [91]. The microbial biomass of these paths increases under the influence of root systems and animal activity. The input of organic matter from preferential flow paths leads to an increase in enzyme activity [28]. Consequently, the accelerated flow within preferential flow paths may facilitate soil carbon decomposition and respiration, thereby influencing carbon cycling rates and soil respiration processes. Furthermore, preferential flow may impact soil water and oxygen status, indirectly affecting the soil carbon cycle. Preferential flow can increase the soil moisture content of specific soil regions, leading to the prolonged wetness of the regions. The presence of preferential flow paths increases soil moisture content and facilitates enhanced water and oxygen availability, thereby providing favorable conditions for the growth and metabolism of soil microorganisms. This augmentation of soil microbial activity can promote organic matter decomposition and carbon conversion processes.

The relationship between preferential flow and soil carbon cycle constitutes a complex field. An in-depth comprehension of the mechanisms underlying the effects of preferential flow on soil carbon transport, transformation, and decomposition is vital for accurate assessments of soil carbon pools, comprehensive understanding of carbon dynamics, and the development of effective soil carbon management strategies. Further research necessitates a combination of field observations, experiments, and model simulations to comprehensively grasp the interactions between preferential flow and the soil carbon cycle.

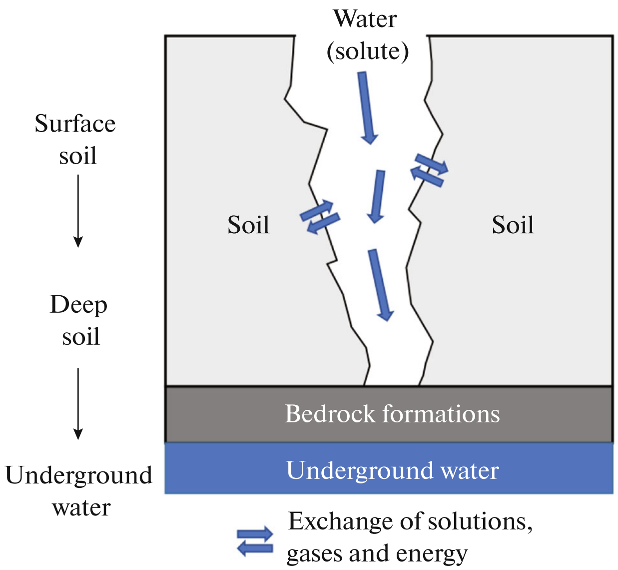

Fig. 3. Rapid non-equilibrium penetration phenomenon of preferential flow.

Preferential Flow Characteristics Assessment and Their Role in Soil Carbon Function

Surrounding characteristics. Preferential flow surrounding characteristics refers to the water flow and solute transport along preferential flow paths which encircle the soil matrix [66]. The surrounding characteristics, to some extent, reflect the complexity of the soil pore networks, which influences the spatial distribution of soil moisture content within the soils. Preferential flow in soil macropores can carry soil dissolved organic matter bypassing the soil matrix from soil surface to the soil deeper layers, increasing the content of soil carbon such layers and causing the heterogeneous spatial distribution of soil carbon increment at different depths [58]. The study of surrounding characteristics is mainly characterized by the dye-stained methods [60]. Stained profile images are processed and analyzed to derive parameters associated with the surrounding characteristics, such as matrix flow depth and stained area ratio [29]. Studies by means of the dye-stained methods demonstrated that water flow could occur in preferential flow paths surrounding the soil matrix. Ohrstrom et al. identified dye-stained paths of different sizes and shapes in the preferential flow paths and applied the sizes and shapes to the description of surrounding characteristics [61]. Furthermore, other methods, such as time-domain reflectometer (TDR), are also used to indicate the distinct phenomenon of surrounding characteristics [14]. The surrounding characteristics provide insights into the complexity of soil water flow processes and are closely related to soil structure health, thus advancing the field of soil physics research and the distribution of soil carbon.

Rapid non-equilibrium penetration characteristics. Rapid non-equilibrium penetration characteristics of preferential flow refers to the phenomenon of fast, non-equilibrium and unstable water flow and solute transport within the soils, which is difficult to be predicted by the adsorption equilibrium theory [62, 71]. In contrast to the surrounding soil matrix, the water flow rate in the preferential flow paths does not reach equilibrium in a short period. At this time, preferential flow carries solutes to penetrate the soil body rapidly to reach the deeper soil layers [39], and the interaction time between solutes and the surrounding soil particles is very short (Fig. 3). Low organic carbon content in preferential flow paths compared to soil matrix leads to the younger soil organic carbon and less humus in preferential flow paths [5]. Using upward osmosis experiments, Šimůnek et al. demonstrated the existence of a non-equilibrium flow phenomenon [72]. Preferential flow paths allow repaid water flow velocity within the soils, and water and solutes penetrate rapidly along these preferential flow paths [17, 34]. The rapid infiltration increases the deeper soil moisture content and promotes the horizontal water flow [45]. Willkommen et al. showed that pesticides can penetrate into the deeper soil layers rapidly [81]. When the solute penetrates rapidly in the preferential flow paths, the solute interacts with the soil particles on the walls of the preferential flow paths. Weakly adsorbed materials in the soil matrix are more readily released from the soil as the intensity of the preferential flow increases [64]. Cote et al found that solutes may diffuse to the soil matrix at the boundary of the preferential flow paths when the flow velocity is low, and solutes are also more likely to leach out when the flow resumes [10]. Such may lead to rapid loss of solutes. In addition, heavy metal content was significantly higher in the preferential flow paths compared to the soil matrix [90]. The preferential flow paths carry solutes (pollutants, pesticides, heavy metals, etc.) through the soil layer rapidly, weakening the strength of the interaction between solutes and the soil matrix, which in turn reduces the efficiency of soil decontamination. The rapid non-equilibrium penetration characteristics promote the study of soil water flow from the homogeneous field to the non-homogeneous research field, which can provide new insights for the non-equilibrium mechanism of soil carbon dynamics.

Fluctuating characteristics. The strength of preferential flow exhibits spatial and temporal differences, demonstrating its fluctuating characteristics (low or high strength of preferential flow). The presence of preferential flow paths play a significant role in accelerating the process of soil carbon cycle. However, the spatial arrangement of preferential flow paths can be fluctuated all the time. These changes pose challenges in monitoring and observing the strength of preferential flow in real-time. In areas with large seasonal differences in precipitation, heavy rainfall during the rainy season promotes an increase in the intensity of preferential flow, which allows soluble organic matter to reach the deeper soil layers and may lead to a decrease in organic matter in surface soils [49]. Wang et al. conducted a study on water infiltration at different time stages, revealing that the fluctuating characteristics of preferential flow might increase in high complexity [80]. Stumpp and Maloszewski investigated preferential flow at different stages of crop growth and showed significant variation in the strength of preferential flow [74]. The dynamic changes in the soil macropore networks, particularly the seasonal fluctuations in the connectivity of macropores, have profound effects on soil carbon dynamics [76, 89]. Changes in the internal soil environment with different seasons affect the nutrient supply to soil organic matter, soil aggregates, and plants [4]. These fluctuations in soil organic matter and soil aggregates, in turn, influence the preferential flow strength [46]. The fluctuating characteristics of preferential flow encompass the multidimensionality, continuity, and prolonged duration of soil carbon dynamics. Thus, the fluctuating characteristics of preferential flow facilitate advancements in soil carbon cycle and represent a concentrated expression of the transient variability in soil carbon dynamics within the soils.

Universal characteristics. Preferential flow is a common process in soils or on the soil surface, highlighting its universal characteristics. In general, preferential flow connects the surface and deeper layers of the soil by common ways (preferential flow paths), such significantly affects soil carbon distribution and microbial activity [19]. Studies have expanded from forests, agricultural land and wetlands to diverse environments, such as tailing mines, desert, and grasslands to indicate the occur of universal phenomenon of preferential flow in different environments. Preferential flows are characterized differently in different types of ecosystems. The development of preferential flows was most favored in natural mixed coniferous and broadleaf forests, and the intensity of preferential flows was lower in broadleaf forests with less ecological diversity [51]. In comparison to forest ecosystems, preferential flows were least favorable in grassland soils [1, 49]. In agricultural soils, preferential flow shows different strength in soils subjected to long-term mechanical tillage and in those under normal tillage practices [85]. Preferential flow evolves from initial fast non-equilibrium flow to slower matrix flow over time [70]. These observations highlight the universality of preferential flow at different soil types or time scales. The widespread occur of preferential flow has led to a consensus in studying the effects of preferential flow on soil carbon dynamics. Moreover, its universality in different conditions contributes to understanding its impact on soil carbon cycle.

Lateral infiltration characteristics. When the soil texture changes, for example, the soil texture changes from loose conditions to more compacted soils, the water flow is impeded and lateral infiltration occurs [69]. Such is treated as preferential flow lateral infiltration characteristics. In general, lateral infiltration characteristics can lead to heterogeneity in soil carbon distribution and horizontal stratification of carbon concentrations across soil layers due to the long-term stability of preferential flow paths [67]. The lateral infiltration promotes the interaction between preferential flow paths and soil matrix. Water flow in soil matrix can be recognized through capillary siphons, while those flow in preferential flow paths by the influence of gravity [23, 83]. During the initial stages of water infiltration, there is the least resistance in the preferential flow paths, and the soil water content within these paths is higher. In contrast, water in small pores or capillary pores moves slowly. As a result, water in the preferential flow paths moved toward the surrounding soil matrix, promoting rapid distribution of water at different soil locations [35]. The lateral infiltration characteristics of preferential flow helps balance water content variations among different soil layers, thereby influencing the efficiency of soil carbon accumulation or transport at different depths.

The Phenomenon of Preferential Flow in Different Ecosystems

The presence of interconnected or disconnected macropore networks and soil fissures contributes to the development of preferential flow [63]. The phenomenon of preferential flow commonly occurs in different ecosystems, and its characteristics vary from ecosystems. A comprehensive understanding of preferential flow effects on soil carbon dynamics in different ecosystems can provide a theoretical basis for further research on preferential flow positive or negative responses.

Farmland ecosystems. Soils in agricultural areas commonly exhibit a layered structure, consisting of a loose tilled layer and a compacted plow bottom layer. When water infiltrates through the loose tilled layers to a less permeable soil layers, it may lead to the formation of preferential flow paths [95] and exhibit lateral infiltration characteristics. A variety of agricultural management practices can affect the intensity of preferential flow and consequently water infiltration into agriculture soil, such as the adoption of conservation tillage methods in arable land [26]. Preferential flow play a crucial role in facilitating the leaching of nitrate-nitrogen. Inadequate application of nitrogen fertilizers and improper irrigation practices can result in the rapid movement of nitrate-nitrogen to deeper soil layers through preferential flow paths. However, the proper distribution of organic and chemical fertilizers along with improved irrigation techniques can reduce nitrogen leaching [50]. Pesticides undergo processes like degradation, volatilization, or adsorption by soil colloids under the influence of physical, chemical, and biological factors. However, due to the universal presence of preferential flow and the rapid non-equilibrium infiltration characteristics of preferential flow, preferential flow present in agricultural soils can carry fertilizers and pesticides for rapid transport and contamination of surface water or groundwater [78]. Therefore, the preferential flow in agricultural soils not only hinders water and nutrient uptake by crops, leading to nutrient wastage, but also increases the cost of agricultural inputs. Moreover, it contributes to environmental issues such as groundwater contamination due to pesticides and fertilizers. In-depth studies on the preferential flow characteristics in agricultural soils can help mitigate the impact of preferential flow paths on pollutant transport caused by fissures resulting from alternating wet and dry conditions and freeze-thaw cycles. This knowledge can provide a theoretical foundation for the efficient utilization of water and nutrients in agricultural practices, as well as the reduction of environmental pollution arising from pesticides and fertilizers.

Forest ecosystems. In comparison to other land use types, forest soils exhibit a higher capacity to mitigate surface runoff, primarily attributed to the extensive networks of preferential flow paths in the forest soils. Forest ecosystems play a vital role in regulating runoff and influencing the quantity and quality of water resources, particularly through the impact of preferential flow on deep water replenishment and vegetation maintenance in forest soils. As a result, these dynamics alter the interaction between forests and water. Various factors in forest ecosystems, including tree species composition, forest age, vegetation density, root distribution, and human forest management, can modify the physical properties of forest soils and the spatial arrangement of roots, consequently influencing the strength of preferential flow [88]. Studies have revealed that water infiltration and preferential flow is stronger in forested areas compared to bare land, agricultural land and scrub [75]. Compared to other ecosystems, the richer root biomass and significant seasonal variation in forested areas have a significant effect on the strength of preferential flow [36, 92]. Specifically, water flow in the preferential flow paths progressively increases as the root system promotes the development of preferential root channels. In sloping soils, the rapid movement of preferential flow into deeper soils and increased water content above the impermeable layer can trigger landslides along the layer. Forest fires lead to the combustion and cementation of organic matter on the soil surface, resulting in the formation of a water-repellent layer that hinders water infiltration [16]. Consequently, rainfall infiltration rates decrease due to the influence of the water-repellent layer, and water is channeled through subsurface soil conduits, rendering the area susceptible to severe soil erosion during heavy rainfall events [42]. Forest ecosystems possess well-structured soils influenced by a thicker layer of surface litter, enhanced microbial activity, and other soil processes, which in turn form an interconnected network of preferential flow paths. This network facilitates rapid movement of soil water and organic carbon to deeper soil layers [52], reduces surface runoff, enhances water retention, and preserves soil and water resources. However, the extensive presence of preferential flow paths within the soil layer can become fully saturated under intense rainfall conditions, significantly increasing gravitational forces on the slope and potentially triggering landslide disasters that impact the functionality of forest vegetation.

Wetland ecosystems. Wetlands possess various functions, including flood regulation, water purification, water source protection, erosion reduction, climate regulation, biodiversity maintenance, and biological productivity provision [44]. The hydrological function of a wetland is greatly dependent on water flow patterns and residence times [38]. Presence of preferential flow paths in wetland soils weakens the interaction between preferential flow and the soil matrix, resulting in less efficient decontamination within wetlands [53]. Under freezing conditions, the rapid infiltration of preferential flow leads to a decrease in the wastewater treatment efficiency of wetlands. Muñoz et al. demonstrated that wetlands located in colder areas can increase vegetative cover, improve aeration, increase internal wetland temperatures, and reduce the occurrence of preferential flow paths within wetland soils [59]. Compared to wetlands without vegetation cover, wetland systems with vegetation cover exhibit stronger solute-soil-vegetation interactions, enhanced denitrification in the preferential flow paths, and higher microbial activity [7, 15]. However, in wetlands with long-term water saturation, the presence of preferential flow paths can enhance oxygen flux and significantly intensify soil redox reactions within the preferential flow paths, thereby promoting wetland vegetation growth [40]. In wetland ecosystems, areas with well-developed plant roots exhibit higher strength of preferential flow, leading to the transportation of solutes along preferential flow paths formed by the root systems and their retention by the channel walls. As a result, the organic carbon, organic matter and nutrient content of the preferential flow paths surpasses that of the soil matrix, significantly influencing nutrient availability within the inter-root zone [82]. The considerable strength of preferential flow in wetland soils affects the water purification capacity of wetlands and the efficiency of pollutant retention, consequently impacting the eco-hydrological function of wetlands. Moreover, the presence of preferential flow paths replenishes wetland soils with ample water, nutrients, and oxygen, thus improving microbial habitats, vegetation growth conditions, and soil productivity within the wetland ecosystems.

Desert ecosystems. The spatial distribution of soil moisture has a significant impact on vegetation distribution and growth, while vegetation, in turn, influences soil moisture distribution [86]. In sandy areas, soil moisture content is usually inversely proportional to the degree of vegetation cover. Different water utilization patterns of deep-rooted shrubs and shallow-rooted plants also affect the spatial distribution of water within the soil profile [48], increasing the amount of water moving deeper into the desert soils through preferential flow paths. The strength of preferential flow gradually increases with larger vegetation root biomass [13]. The preferential flow paths facilitates the replenishment of soil water in deeper layers, creating favorable soil water conditions for vegetation growth in arid desert environments [47]. Additionally, preferential flow development under shrub vegetation enriches soil nutrients [47]. Agricultural farming or afforestation in desert areas requires substantial improvements in soil water storage, which further complicates water transfer dynamics [94]. Biological crusts on sandy surfaces can enhance sandy soil structure, but they also impede water movement into deeper soils [84], resulting in reduced water infiltration rates and discouraging the growth of deep-rooted vegetation. In the restored area of Maowusu sand, shallow-rooted vegetation gradually dominates, contributing to the development of surface crust. This, in turn, decreases the soil water infiltration capacity and increases hydrological connectivity on the soil crust surface, influencing water redistribution and the ecological succession process in the sand [43]. Therefore, studying preferential flow in desert areas can elucidate the causes of severe degradation in sandy regions, provide data support for vegetation and ecological restoration in desert areas, and offer decision-making references for the efficient utilization of limited water resources in the region.

Permafrost ecosystems. The infiltration capacity of soil water in permafrost regions plays a crucial role in regulating the distribution of meltwater between infiltration and runoff, thereby significantly impacting the soil water movement process in permafrost ecosystems. Snowmelt serves as the primary source of groundwater recharge in cold regions, freeze-thaw cycles induce changes in soil physical properties and facilitate the formation of soil macropores, which enhance water movement and increase infiltration capacity [56], consequently influencing the hydrothermal distribution within the soils. Preferential flow occurs in both vertically and lateral soil layer of permafrost soils. In cases where snow and ice melt on slopes while the deeper soil remains frozen, infiltrated water undergoes refreezing, resulting in reduced soil water infiltration and the generation of surface runoff. This phenomenon weakening the exchange processes with the frozen soil matrix, thereby diminishing the soil's filtration capacity [57]. Soils in cold regions are affected by freeze-thaw, which affects the soil microstructure and leading to the formation of soil cracks and increased soil permeability, and reducing soil stability. During soil freezing, water exchange channels between the frozen layer and the underlying unfrozen layer become isolated, resulting in the accumulation of soil moisture in the frozen layer. Consequently, the strength of slope soil decreases and the risk of natural disasters increases, such as landslides upon thawing of the frozen layer [11]. Therefore, the study of preferential flow in permafrost ecosystems represents a current research hotspot and a challenging area. Despite the limitations and unique characteristics of natural conditions in these regions, investigating preferential flow can help mitigate soil instability, unravel the mechanisms of preferential water movement due to soil thawing under alternating cold and warm conditions in permafrost areas, and thus contribute to enriching the understanding of preferential flow in specialized ecosystems.

Perspectives of Preferential Flow Research

This paper provides a comprehensive exploration of the distinct characteristics of preferential flow and elucidates its prominent roles in ecosystems. Despite the abundance of research reports on preferential flow both domestically and internationally [93], accurately quantifying and analyzing the preferential flow process remains challenging. Given the current state of preferential flow research, it is imperative to prioritize the following areas for further investigation:

(1) Current knowledge underscores the pivotal role of preferential flow in soil carbon cycling; however, numerous questions remain unexplored. The current research predominantly revolves around the analysis of how preferential flow affects soil carbon loss and distribution, providing preliminary insights. However, the intricate mechanisms through which preferential flow characteristics, including rapid non-equilibrium characteristics, fluctuation characteristics, and lateral infiltration characteristics, affect the dynamics of soil carbon necessitate further investigation, necessitating tailored research approaches. Moreover, the indirect effects of preferential flow on soil carbon warrant further exploration, including its influence on microbiota, root activity, soil organic matter decomposition, and stability, all of which play critical roles in soil carbon cycling. Therefore, we advocate for comprehensive quantitative investigations into the interplay between preferential flow and soil carbon. This approach will augment our comprehension of how preferential flow influences soil carbon cycling, thus furnishing robust scientific support for sustainable land management and carbon sequestration.

(2) Preferential flow is widespread in soils, and this review summarizes five characteristics of preferential flow, which are surrounding characteristics, rapid non-equilibrium infiltration characteristics, fluctuating characteristics, universal characteristics, lateral infiltration characteristics. However, due to the complexity of preferential flow, its characteristic manifestations need to be explored more deeply. For example, the bacterial and microbial activities during water flow inside the soils may produce some secretions to block the channels and affect the flow of water [79].

(3) This review provides a comprehensive review of the phenomenon of preferential flow in agricultural, forests, wetlands, deserts, and permafrost ecosystems. However, there are additional ecosystem types with preferential flow features that warrant attention, such as some large-scale disused soil disposal sites, where soil structure and water movement characteristics differ from those described in the aforementioned five ecosystem types. The study of preferential flow in such ecosystems deserves special consideration.

Furthermore, in addition to these three main issues, common but intractable problems at this stage of preferential flow research need to be given more attention.

(4) Soil macropores are crucial contributors to preferential flow formation. Changes in soil macropores induced by vegetation roots, soil biological activities, and environmental fluctuations, which cause preferential flow to exhibit a variety of properties, travelling rapidly along preferential pathways and encircling portions of the soil matrix. The spatial distribution of preferential flow is usually studied using the dye tracer method, but this method makes it difficult to accurately analyze the three-dimensional spatial and temporal distribution characteristics of preferential flow paths. Recent advancements in NMR, X-ray and CT scanning techniques have facilitated the quantification of the 3D structure of root systems and macropores within the soils [22]. Techniques such as fluorescent staining colloids for labeling preferential flow paths and CT scanning for obtaining 3D pore networks have been employed to determine preferential flow paths in soils [73]. However, further studies are required to investigate the spatial structural features that exert more significant influences on hydrological processes.

(5) The non-equilibrium and heterogeneous nature of preferential flow in soils hinders a comprehensive understanding of the underlying mechanisms. and preferential flows are prevalent in a wide range of soils. Geographical variations in preferential flow studies result from diverse factors. Future research on preferential flow should focus on collecting and integrating geographic, climatic, vegetation, and soil data from different study areas. Additionally, sophisticated detection methods should be employed to examine preferential flow characteristics. For instance, real-time monitoring of soil water content changes using TDR arrays can provide insights into the dynamic changes in preferential flow [76]. Analyzing the correlation between various factors influencing preferential flow and their effects with approaches like artificial neural networks and sensitivity analysis is crucial. These advancements will contribute to comparative studies of preferential flow in different regions and facilitate regional water resource management and regulation.

(6) Preferential flow infiltrates rapidly, bypassing the soil matrix, infiltrating laterally between different soil layers, and playing a distinct role in water transport and nutrient delivery. The extensive networks formed by interconnected preferential flow paths in the soil exhibit temporal stability and can remain active for periods [25], resulting in elevated concentrations of carbon and microbial biomass. Exploring soil carbon cycling, soil microbial ecology, and geochemical processes within preferential flow networks can provide valuable insights into the impact of these networks on soil functions [19]. For instance, in forest ecosystems, preferential flow can contribute to deep soil water replenishment and nutrient cycling. The "water-energy-food-forest" system can be fully taken into account in studies related to the preferential flow of forests [54] to quantify the importance and contribution of the preferential flow to the sustainable development of this system. By addressing these research, further advancements in understanding the complex preferential flow will be achieved, providing valuable insights into its role in various ecosystems and enabling effective management of water resources.

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AUTHOR CONTRIBUTION

Conceptualization, Yinghu Zhang; investigation, Wenqi Zhang, Lu Wang Jinhong Chen; writing, Wenqi Zhang, review and editing, Yinghu Zhang, Wenqi Zhang; funding acquisition, Yinghu Zhang. All authors have read and agreed to the published version of the manuscript.

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ETHICS APPROVAL AND CONSENT TO PARTICIPATE

This work does not contain any studies involving human and animal subjects.

CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

REFERENCES

1. A. Alaoui, U. Caduff, H. H. Gerke, and R. Weingartner, "Apreferential flow effects on infiltration and runoff in grassland and forest soils," Vadose Zone J. **10** (1), 367–377 (2011). https://doi.org/10.2136/vzj2010.0076

2. S. E. Allaire, S. Roulier, and A. J. Cessna, "Quantifying preferential flow in soils: a review of different techniques," J. Hydrol. **378** (1–2), 179–204 (2009). https://doi.org/10.1016/j.jhydrol.2009.08.013
3. V. Bailey, A. Smith, M. Tfaily, S. Fansler, and B. Bond-Lamberty, "Differences in soluble organic carbon chemistry in pore waters sampled from different pore size domains," Soil Biol. Biochem. **107**, 133–143 (2017). https://doi.org/10.1016/j.soilbio.2016.11.025
4. L. Brussaard, M. M. Pulleman, É. Ouédraogo, A. Mando, and J. Six, "Soil fauna and soil function in the fabric of the food web," Pedobiologia **50** (6), 447–462 (2007). https://doi.org/10.1016/j.pedobi.2006.10.007
5. M. Bundt, M. Jäggi, P. Blaser, R. Siegwolf, and F. Hagedorn, "Carbon and nitrogen dynamics in preferential flow paths and matrix of a forest soil," Soil Sci. Soc. Am. J. **65** (5), 1529–1538 (2001). https://doi.org/10.2136/sssaj2001.6551529x
6. M. Bundt, F. Widmer, M. Pesaro, J. Zeyer, and P. Blaser, "Preferential flow paths: biological 'hot spots' in soils," Soil Biol. Biochem. **33** (6), 729–738 (2001). https://doi.org/10.1016/S0038-0717(00)00218-2
7. R. E. Casey, M. Taylor, and S. J. Klaine, "Localization of denitrification activity in macropores of a riparian wetland," Soil Biol. Biochem. **36** (4), 563–569 (2004). https://doi.org/10.1016/j.soilbio.2003.11.003
8. A. Chabbi, I. Kögel-Knabner, and C. Rumpel, "Stabilised carbon in subsoil horizons is located in spatially distinct parts of the soil profile," Soil Biol. Biochem. **41** (2), 256–261 (2009). https://doi.org/10.1016/j.soilbio.2008.10.033
9. B. Clothier, S. Green, and M. Deurer, "Preferential flow and transport in soil: progress and prognosis," Eur. J. Soil Sci. **59** (1), 2–13 (2008). https://doi.org/10.1111/j.1365-2389.2007.00991.x
10. C. Cote, K. L. Bristow, and P. Ross, "Increasing the efficiency of solute leaching: impacts of flow interruption with drainage of the "preferential flow paths"," J. Contam. Hydrol. **43** (3–4), 191–209 (2000). https://doi.org/10.1016/s0169-7722(00)00087-5
11. K. Cui and X. Qin, "Landslide risk assessment of frozen soil slope in Qinghai Tibet Plateau during spring thawing period under the coupling effect of moisture and heat," Nat. Hazards **115** (3), 2399–2416 (2023). https://doi.org/10.1007/s11069-022-05646-8
12. T. H. Debieche, T. Bogaard, V. Marc, C. Emblanch, D. Krzeminska, and J. P. Malet, "Hydrological and hydrochemical processes observed during a large-scale infiltration experiment at the Super-Sauze mudslide (France)," Hydrol. Processes **26** (14), 2157–2170 (2012). https://doi.org/10.1002/hyp.7843
13. D. A. Devitt and S. D. Smith, "Root channel macropores enhance downward movement of water in a Mojave Desert ecosystem," J. Arid Environ. **50** (1), 99–108 (2002). https://doi.org/10.1006/jare.2001.0853
14. X. Duan, C. Ni, J. Chen, and J. Chen, "Study on the preferential flow of red soil erosion in granite slope collapse with high frequency monitoring of water content," J. Soil Water Conserv. **30**, 82–88 (2016). https://doi.org/10.13870/j.cnki.stbcxb.2016.05.014
15. R. Durst, G. Imfeld, and J. Lange, "Transport of pesticides and artificial tracers in vertical-flow lab-scale wetlands," Water Resour. Res. **49** (1), 554–564 (2013). https://doi.org/10.1002/wrcr.20080
16. A. Dymov, "Soils of post-pyrogenic forests," Eurasian Soil Sci. **56** (Suppl. 1), S84–S113 (2023). https://doi.org/10.1134/S1064229323700217
17. M. Flury, H. Flühler, W. A. Jury, and J. Leuenberger, "Susceptibility of soils to preferential flow of water: a field study," Water Resour. Res. **30** (7), 1945–1954 (1994). https://doi.org/10.1029/94wr00871
18. S. Franklin, B. Vasilas, and Y. Jin, "More than meets the dye: evaluating preferential flow paths as microbial hotspots," Vadose Zone J. **18** (1), 1–8 (2019). https://doi.org/10.2136/vzj2019.03.0024
19. S. M. Franklin, A. N. Kravchenko, R. Vargas, B. Vasilas, J. J. Fuhrmann, and Y. Jin, "The unexplored role of preferential flow in soil carbon dynamics," Soil Biol. Biochem. **161**, 108398 (2021). https://doi.org/10.1016/j.soilbio.2021.108398
20. I. Fuhrmann, S. Maarastawi, J. Neumann, W. Amelung, K. Frindte, C. Knief, E. Lehndorff, R. Wassmann, and J. Siemens, "Preferential flow pathways in paddy rice soils as hot spots for nutrient cycling," Geoderma **337**, 594–606 (2019). https://doi.org/10.1016/j.geoderma.2018.10.011
21. F. Garrido, S. Serrano, L. Barrios, J. Uruñuela, and M. Helmhart, "Preferential flow and metal distribution in a contaminated alluvial soil from São Domingos mine (Portugal)," Geoderma **213**, 103–114 (2014). https://doi.org/10.1016/j.geoderma.2013.07.034
22. K. Gerke, E. Skvortsova, and D. Korost, "Tomographic method of studying soil pore space: Current perspectives and results for some Russian soils," Eurasian Soil Sci. **45**, 700–709 (2012). https://doi.org/10.1134/S1064229312070034
23. P. F. Germann and K. Beven, "Kinematic wave approximation to infiltration into soils with sorbing macropores," Water Resour. Res. **21** (7), 990–996 (1985). https://doi.org/10.1029/wr021i007p00990
24. F. Hagedorn, N. Bruderhofer, A. Ferrari, and P. A. Niklaus, "Tracking litter-derived dissolved organic matter along a soil chronosequence using 14C imaging: biodegradation, physico-chemical retention or preferential flow?," Soil Biol. Biochem. **88**, 333–343 (2015). https://doi.org/10.1016/j.soilbio.2015.06.014
25. F. Hagedorn and M. Bundt, "The age of preferential flow paths," Geoderma **108** (1–2), 119–132 (2002). https://doi.org/10.1016/s0016-7061(02)00129-5
26. E. Hangen, U. Buczko, O. Bens, J. Brunotte, and R. Hüttl, "Infiltration patterns into two soils under conventional and conservation tillage: influence of the spatial distribution of plant root structures and soil animal activity," Soil Tillage Res. **63** (3–4), 181–186 (2002). https://doi.org/10.1016/s0167-1987(01)00234-3
27. D. R. Hirmas, D. Giménez, A. Nemes, R. Kerry, N. A. Brunsell, and C. J. Wilson, "Climate-induced changes in continental-scale soil macroporosity may intensify water cycle," Nature **561** (7721), 100–103 (2018). https://doi.org/10.1038/s41586-018-0463-x

28. D. T. Hoang, J. Pausch, B. S. Razavi, I. Kuzyakova, C. C. Banfield, and Y. Kuzyakov, "Hotspots of microbial activity induced by earthworm burrows, old root channels, and their combination in subsoil," Biol. Fertil. Soils **52**, 1105–1119 (2016). https://doi.org/10.1007/s00374-016-1148-y

29. F. Hou, J. Cheng, and N. Guan, "Investigating the effect of soil cracks on preferential flow using a dye tracing infiltration experiment in karst in Southwest China," Land Degrad. Dev. **34** (6), 1612–1628 (2023). https://doi.org/10.1002/ldr.4557

30. L. Jačka, A. Walmsley, M. Kovář, and J. Frouz, "Effects of different tree species on infiltration and preferential flow in soils developing at a clayey spoil heap," Geoderma **403**, 115372 (2021). https://doi.org/10.1016/j.geoderma.2021.115372

31. N. Jarvis, "A review of non-equilibrium water flow and solute transport in soil macropores: principles, controlling factors and consequences for water quality," Eur. J. Soil Sci. **71** (3), 279–302 (2020). https://doi.org/10.1111/ejss.12973

32. N. Jarvis, J. Koestel, and M. Larsbo, "Understanding preferential flow in the vadose zone: Recent advances and future prospects," Vadose Zone J. **15** (12), 1–11 (2016). https://doi.org/10.2136/vzj2016.09.0075

33. P. Jiang, L. Xiao, X. Wan, T. Yu, Y. Liu, and M. Liu, "Research progress on microbial carbon sequestration in soil: a review," Eurasian Soil Sci. **55** (10), 1395–1404 (2022). https://doi.org/10.1134/S1064229322100064

34. X.-j. Jiang, E.-h. Wang, X.-w. Chen, X.-y. Xia, and C.-t. Shi, "Surrounding characteristics of preferential flow in cultivated typical black soils of Northeast China," Yingyong Shengtai Xuebao **21** (12) (2010). https://sci-hub.wf/10.13287/j.1001-9332.2010.0444

35. X.-J. Jiang, S. Zakari, J. Wu, A. K. Singh, C. Chen, X. Zhu, W. Zhang, and W. Liu, "Can complementary preferential flow and non-preferential flow domains contribute to soil water supply for rubber plantation?," For. Ecol. Manage. **461**, 117948 (2020). https://doi.org/10.1016/j.foreco.2020.117948

36. X. Kan, J. Cheng, X. Hu, F. Zhu, and M. Li, "Effects of grass and forests and the infiltration amount on preferential flow in karst regions of China," Water **11** (8), 1634 (2019). https://doi.org/10.3390/w11081634

37. S. Keleş, "An assessment of hydrological functions of forest ecosystems to support sustainable forest management," J. Sustainable For. **38** (4), 305–326 (2019). https://doi.org/10.1080/10549811.2018.1547879

38. J. Kjellin, A. Wörman, H. Johansson, and A. Lindahl, "Controlling factors for water residence time and flow patterns in Ekeby treatment wetland, Sweden," Adv. Water Resour. **30** (4), 838–850 (2007). https://doi.org/10.1016/j.advwatres.2006.07.002

39. K.-J. Kung, T. Steenhuis, E. Kladivko, T. Gish, G. Bubenzer, and C. Helling, "Impact of preferential flow on the transport of adsorbing and non-adsorbing tracers," Soil Sci. Soc. Am. J. **64** (4), 1290–1296 (2000). https://doi.org/10.2136/sssaj2000.6441290x

40. C. Lahiri and G. R. Davidson, "Heterogeneous oxygenation of wetland soils with increasing inundation: redox potential, water depth, and preferential flow paths," Hydrol. Processes **34** (6), 1350–1358 (2020). https://doi.org/10.1002/hyp.13654

41. A. Legout, C. Legout, C. Nys, and E. Dambrine, "Preferential flow and slow convective chloride transport through the soil of a forested landscape (Fougères, France)," Geoderma **151** (3–4), 179–190 (2009). https://doi.org/10.1016/j.geoderma.2009.04.002

42. J. Letey, "Causes and consequences of fire-induced soil water repellency," Hydrol. Processes **15** (15), 2867–2875 (2001). https://doi.org/10.1002/hyp.378

43. B. Li, J. Gao, X. Wang, L. Ma, Q. Cui, and M. Vest, "Effects of biological soil crusts on water infiltration and evaporation Yanchi Ningxia, Maowusu Desert, China," Int. J. Sediment Res. **31** (4), 311–323 (2016). https://doi.org/10.1016/j.ijsrc.2016.05.005

44. J. Li, Q. Chen, Q. Li, C. Zhao, and Y. Feng, "Influence of plants and environmental variables on the diversity of soil microbial communities in the Yellow River Delta Wetland, China," Chemosphere **274**, 129967 (2021). https://doi.org/10.1016/j.chemosphere.2021.129967

45. M. Li, J. Yao, R. Yan, and J. Cheng, "Effects of infiltration amounts on preferential flow characteristics and solute transport in the protection forest soil of southwestern China," Water **13** (9), 1301 (2021). https://doi.org/10.3390/w13091301

46. T. Li, M. a. Shao, Y. Jia, X. Jia, and L. Huang, "Small-scale observation on the effects of the burrowing activities of mole crickets on soil erosion and hydrologic processes," Agric., Ecosyst. Environ. **261**, 136–143 (2018). https://doi.org/10.1016/j.agee.2018.04.010

47. X.-Y. Li, Z.-P. Yang, Y.-T. Li, and H. Lin, "Connecting ecohydrology and hydropedology in desert shrubs: stemflow as a source of preferential flow in soils," Hydrol. Earth Syst. Sci. **13** (7), 1133–1144 (2009). https://doi.org/10.5194/hess-13-1133-2009

48. X. R. Li, F. Y. Ma, H. L. Xiao, X. P. Wang, and K. C. Kim, "Long-term effects of revegetation on soil water content of sand dunes in arid region of Northern China," J. Arid Environ. **57** (1), 1–16 (2004). https://doi.org/10.1016/S0140-1963(03)00089-2

49. C. M. T. Lins, E. R. de Souza, T. E. M. dos Santos Souza, M. K. S. S. Paulino, D. R. Monteiro, V. S. de Souza Júnior, P. R. M. Dourado, F. E. d. A. R. Junior, Y. J. A. da Silva, and B. Schaffer, "Influence of vegetation cover and rainfall intensity on soil attributes in an area undergoing desertification in Brazil," Catena **221**, 106751 (2023). https://doi.org/10.1016/j.catena.2022.106751

50. W. Liu, G. Ma, C. Wang, J. Wang, H. Lu, S. Li, W. Feng, Y. Xie, D. Ma, and G. Kang, "Irrigation and nitrogen regimes promote the use of soil water and nitrate nitrogen from deep soil layers by regulating root growth in wheat," Front. Plant Sci. **9**, 32 (2018). https://doi.org/10.3389/fpls.2018.00032

51. Z. Luo, J. Niu, L. Zhang, X. Chen, W. Zhang, B. Xie, J. Du, Z. Zhu, S. Wu, and X. Li, "Roots-enhanced preferential flows in deciduous and coniferous Forest soils revealed by dual-tracer experiments," J. Environ.

Qual. **48** (1), 136–146 (2019).
https://doi.org/10.2134/jeq2018.03.0091

52. E. Marin-Spiotta, O. A. Chadwick, M. Kramer, and M. S. Carbone, "Carbon delivery to deep mineral horizons in Hawaiian rain forest soils," J. Geophys. Res.: Biogeosci. **116** (G3) (2011).
https://doi.org/10.1029/2010JG001587
53. G. McCarty and J. Angier, presented at the *Preferential Flow: Water Movement and Chemical Transport in the Environment* (2001) (unpublished).
54. F. P. Melo, L. Parry, P. H. Brancalion, S. R. Pinto, J. Freitas, A. P. Manhães, P. Meli, G. Ganade, and R. L. Chazdon, "Adding forests to the water–energy–food nexus," Nat. Sustainability **4** (2), 85–92 (2021).
https://doi.org/10.1038/s41893-020-00608-z
55. G. Miguez-Macho and Y. Fan, "The role of groundwater in the Amazon water cycle: 2. Influence on seasonal soil moisture and evapotranspiration," J. Geophys. Res.: Atmos. **117** (D15) (2012).
https://doi.org/10.1029/2012JD017539
56. A. A. Mohammed, E. E. Cey, M. Hayashi, M. V. Callaghan, Y. J. Park, K. L. Miller, and S. K. Frey, "Dual-permeability modeling of preferential flow and snowmelt partitioning in frozen soils," Vadose Zone J. **20** (2), e20101 (2021).
https://doi.org/10.1002/vzj2.20101
57. A. A. Mohammed, I. Pavlovskii, E. E. Cey, and M. Hayashi, "Effects of preferential flow on snowmelt partitioning and groundwater recharge in frozen soils," Hydrol. Earth Syst. Sci. **23** (12), 5017–5031 (2019).
https://doi.org/10.5194/hess-23-5017-2019
58. Y. Mori, A. Fujihara, and K. Yamagishi, "Installing artificial macropores in degraded soils to enhance vertical infiltration and increase soil carbon content," Prog. Earth Planet. Sci. **1** (1), 1–10 (2014).
https://doi.org/10.1186/s40645-014-0030-5
59. P. Munoz, A. Drizo, and W. C. Hession, "Flow patterns of dairy wastewater constructed wetlands in a cold climate," Water Res. **40** (17), 3209–3218 (2006).
https://doi.org/10.1016/j.watres.2006.06.036
60. S. Ogawa, P. Baveye, C. W. Boast, J.-Y. Parlange, and T. Steenhuis, "Surface fractal characteristics of preferential flow patterns in field soils: evaluation and effect of image processing," Geoderma **88** (3–4), 109–136 (1999).
https://doi.org/10.1016/s0016-7061(98)00101-3
61. P. Öhrström, M. Persson, J. Albergel, P. Zante, S. Nasri, R. Berndtsson, and J. Olsson, "Field-scale variation of preferential flow as indicated from dye coverage," J. Hydrol. **257** (1–4), 164–173 (2002).
https://doi.org/10.1016/s0022-1694(01)00537-6
62. Z. On, L. Jia, H. Jin, X. Jiang, Q. Zhang, and J. Gao, "Macropores and preferential flow and their effects on pollutant migration in soils," Acta Pedol. Sin. **36** (3), 341–347 (1999).
https://doi.org/10.11766/trxb199802240307
63. F. Preti, E. Guastini, D. Penna, A. Dani, G. Cassiani, J. Boaga, R. Deiana, N. Romano, P. Nasta, and M. Palladino, "Conceptualization of water flow pathways in agricultural terraced landscapes," Land Degrad. Dev. **29** (3), 651–662 (2018).
https://doi.org/10.1002/ldr.2764
64. J. Radolinski, H. Le, S. S. Hilaire, K. Xia, D. Scott, and R. D. Stewart, "A spectrum of preferential flow alters solute mobility in soils," Sci. Rep. **12** (1), 4261 (2022).
https://doi.org/10.1038/s41598-022-08241-w
65. R. Ragab and C. Prudhomme, "Sw—soil and water: climate change and water resources management in arid and semi-arid regions: prospective and challenges for the 21st century," Biosyst. Eng. **81** (1), 3–34 (2002).
https://doi.org/10.1006/bioe.2001.0013
66. G.-L. Ren, B. Izadi, B. King, and E. Dowding, "Preferential transport of bromide in undisturbed cores under different irrigation methods," Soil Sci. **161** (4), 214–225 (1996).
https://doi.org/10.1097/00010694-199604000-00002
67. C. Rumpel and I. Kögel-Knabner, "Deep soil organic matter—a key but poorly understood component of terrestrial C cycle," Plant Soil **338**, 143–158 (2011).
https://doi.org/10.1007/s11104-010-0391-5
68. C. Rye and K. Smettem, "The effect of water repellent soil surface layers on preferential flow and bare soil evaporation," Geoderma **289**, 142–149 (2017).
https://doi.org/10.1016/j.geoderma.2016.11.032
69. E. Shein, D. Shcheglov, A. Umarova, I. Sokolova, and E. Y. Milanovskii, "Structural status of technogenic soils and the development of preferential water flows," Eurasian Soil Sci. **42**, 636–644 (2009).
https://doi.org/10.1134/S1064229309060088
70. F. Sheng, K. Wang, R. Zhang, and H. Liu, "Modeling the heterogeneous soil water flow and solute transport by two-region-two-stage model," J. Hydraul. Eng. **46** (4), 433–442 (2015).
https://doi.org/10.13243/j.cnki.slxb.2015.04.007
71. F. Sheng, L. Zhang, and D. Wu, "Review on research theories and observation techniques for preferential flow in unsaturated soil," Trans. Chin. Soc. Agric. Eng. **32** (6), 1–10 (2016).
https://doi.org/10.11975/j.issn.1002-6819.2016.06.001
72. J. Šimůnek, O. Wendroth, N. Wypler, and M. T. Van Genuchten, "Non-equilibrium water flow characterized by means of upward infiltration experiments," Eur. J. Soil Sci. **52** (1), 13–24 (2001).
https://doi.org/10.1046/j.1365-2389.2001.00361.x
73. D. Soto-Gómez, P. Pérez-Rodríguez, L. Vazquez Juiz, J. E. López-Periago, and M. Paradelo Perez, "A new method to trace colloid transport pathways in macroporous soils using X-ray computed tomography and fluorescence macrophotography," Eur. J. Soil Sci. **70** (3), 431–442 (2019).
https://doi.org/10.1111/ejss.12783
74. C. Stumpp and P. Maloszewski, "Quantification of preferential flow and flow heterogeneities in an unsaturated soil planted with different crops using the environmental isotope δ18O," J. Hydrol. **394** (3–4), 407–415 (2010).
https://doi.org/10.1016/j.jhydrol.2010.09.014
75. D. Sun, H. Yang, D. Guan, M. Yang, J. Wu, F. Yuan, C. Jin, A. Wang, and Y. Zhang, "The effects of land use change on soil infiltration capacity in China: a meta-analysis," Sci. Total Environ. **626**, 1394–1401 (2018).
https://doi.org/10.1016/j.scitotenv.2018.01.104

76. K. Taumer, H. Stoffregen, and G. Wessolek, "Seasonal dynamics of preferential flow in a water repellent soil," Vadose Zone J. **5** (1), 405–411 (2006). https://doi.org/10.2136/vzj2005.0031

77. G. Van Der Heijden, A. Legout, B. Pollier, C. Bréchet, J. Ranger, and E. Dambrine, "Tracing and modeling preferential flow in a forest soil—potential impact on nutrient leaching," Geoderma **195**, 12–22 (2013). https://doi.org/10.1016/j.geoderma.2012.11.004

78. M. Villamizar and C. D. Brown, "A modelling framework to simulate river flow and pesticide loss via preferential flow at the catchment scale," Catena **149**, 120–130 (2017). https://doi.org/10.1016/j.catena.2016.09.009

79. H. Wang, J. Xin, X. Zheng, M. Li, Y. Fang, and T. Zheng, "Clogging evolution in porous media under the coexistence of suspended particles and bacteria: insights into the mechanisms and implications for groundwater recharge," J. Hydrol. **582**, 124554 (2020). https://doi.org/10.1016/j.jhydrol.2020.124554

80. Z. Wang, L. Wu, T. Harter, J. Lu, and W. A. Jury, "A field study of unstable preferential flow during soil water redistribution," Water Resour. Res. **39** (4), (2003). https://doi.org/10.1029/2001wr000903

81. S. Willkommen, J. Lange, U. Ulrich, M. Pfannerstill, and N. Fohrer, "Field insights into leaching and transformation of pesticides and fluorescent tracers in agricultural soil," Sci. Total Environ. **751**, 141658 (2021). https://doi.org/10.1016/j.scitotenv.2020.141658

82. Y. Wu, Y. Zhang, L. Xie, S. Zhao, Y. Liu, and Z. Zhang, "Preferential flow improves root-soil system on a small scale: A case study of two ecotypes of Phragmites communis," J. Cleaner Prod. **328**, 129581 (2021). https://doi.org/10.1016/j.jclepro.2021.129581

83. Y. Wu, Y. Zhang, Z. Zhenming, and M. Zhang, "Differences in soil characteristics between preferential and matrix flow areas in wetland plants root zone in the Yellow River Delta," J. Beijing Norm. Univ.(Nat. Sci.)**57** (1), 69–75 (2021). https://doi.org/10.12202/j.0476-0301.2020435

84. A. Yair, R. Almog, and M. Veste, "Differential hydrological response of biological topsoil crusts along a rainfall gradient in a sandy arid area: Northern Negev desert, Israel," Catena **87** (3), 326–333 (2011). https://doi.org/10.1016/j.catena.2011.06.015

85. J.-l. Yan and W.-z. Zhao, "Effects of long-term mechanical compaction on preferential flow in oasis cropland," Chin. J. Ecol. **38** (5), 1376 (2019). https://doi.org/10.13292/j.1000-4890.201905.001

86. L. Yang, L. Chen, and W. Wei, "Effects of vegetation restoration on the spatial distribution of soil moisture at the hillslope scale in semi-arid regions," Catena **124**, 138–146 (2015). https://doi.org/10.1016/j.catena.2014.09.014

87. S.-H. Yao, B. Zhang, and F. Hu, "Soil biophysical controls over rice straw decomposition and sequestration in soil: the effects of drying intensity and frequency of drying and wetting cycles," Soil Biol. Biochem. **43** (3), 590–599 (2011). https://doi.org/10.1016/j.soilbio.2010.11.027

88. J. Yi, Y. Yang, M. Liu, W. Hu, S. Lou, H. Zhang, and D. Zhang, "Characterising macropores and preferential flow of mountainous forest soils with contrasting human disturbances," Soil Res. **57** (6), 601–614 (2019). https://doi.org/10.1071/sr18198

89. J. Zhang, H. Lin, and J. Doolittle, "Soil layering and preferential flow impacts on seasonal changes of GPR signals in two contrasting soils," Geoderma **213**, 560–569 (2014). https://doi.org/10.1016/j.geoderma.2013.08.035

90. W. Zhang, F. Jiang, and W. Sun, "Investigating colloid-associated transport of cadmium and lead in a clayey soil under preferential flow conditions," Water Sci. Technol. **84** (9), 2486–2498 (2021). https://doi.org/10.2166/wst.2021.441

91. W. Zhang, L. Wang, Z. Tang, and Y. Zhang, "Effects of the root system architecture of Pinus taeda and Phyllostachys edulis on the index of hydrological connectivity in subtropical forest ecosystems," Forests **13** (12), 2008 (2022). https://doi.org/10.3390/f13122008

92. Y. Zhang, J. Niu, M. Zhang, Z. Xiao, and W. Zhu, "Interaction between plant roots and soil water flow in response to preferential flow paths in northern China," Land Degrad. Dev. **28** (2), 648–663 (2017). https://doi.org/10.1002/ldr.2592

93. Y. Zhang, M. Zhang, J. Niu, and H. Zheng, "The preferential flow of soil: A widespread phenomenon in pedological perspectives," Eurasian Soil Sci. **49**, 661–672 (2016). https://doi.org/10.1134/S1064229316060120

94. Y. Zhang, W. Zhao, X. Li, A. Jia, and W. Kang, "Contribution of soil macropores to water infiltration across different land use types in a desert–oasis ecoregion," Land Degrad. Dev. **32** (4), 1751–1760 (2021). https://doi.org/10.1002/ldr.3823

95. Z. Zhang, H. Zhou, Q. Zhao, H. Lin, and X. Peng, "Characteristics of cracks in two paddy soils and their impacts on preferential flow," Geoderma **228**, 114–121 (2014). https://doi.org/10.1016/j.geoderma.2013.07.026

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